

Gradient descent

Foundations of AI

Carlos Cotrini

UZH CAS Gamechanger AI

26 September 2026

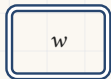
1

Section 1

From the loop to a curve

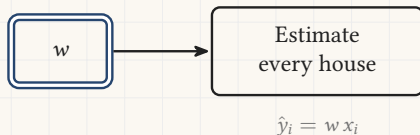
One round, as a function

One turn of the loop from the first lecture, with the machinery boxed up.



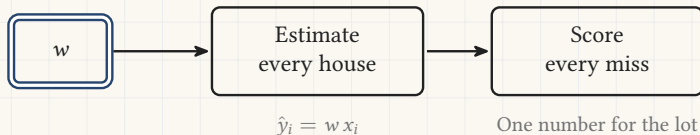
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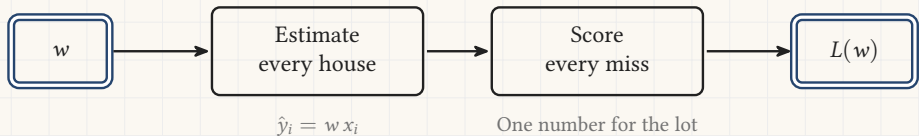
One round, as a function

One turn of the loop from the first lecture, with the machinery boxed up.



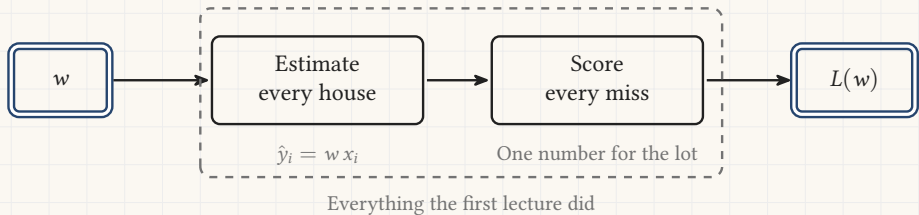
One round, as a function

One turn of the loop from the first lecture, with the machinery boxed up.



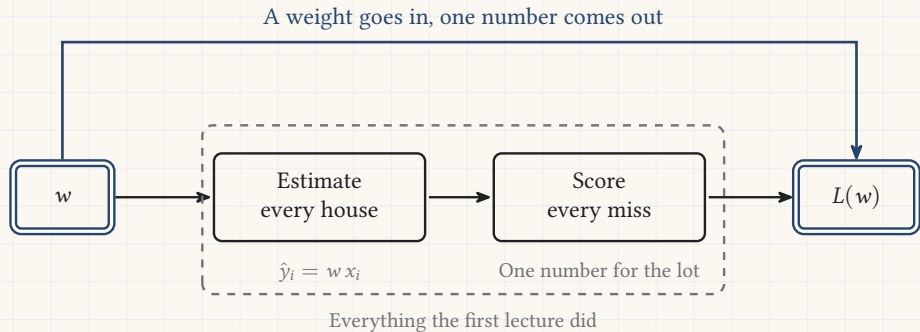
One round, as a function

One turn of the loop from the first lecture, with the machinery boxed up.



One round, as a function

One turn of the loop from the first lecture, with the machinery boxed up.



Scoring a guess

Two flats, both 100 square metres, priced differently. One guess: $w = 25\,000$ per square metre.

Flat	Price	Estimate	Miss

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The average miss

$$\frac{(-500\,000) + (+500\,000)}{2} = 0$$

Zero, and both flats are half a million out.

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The average squared miss

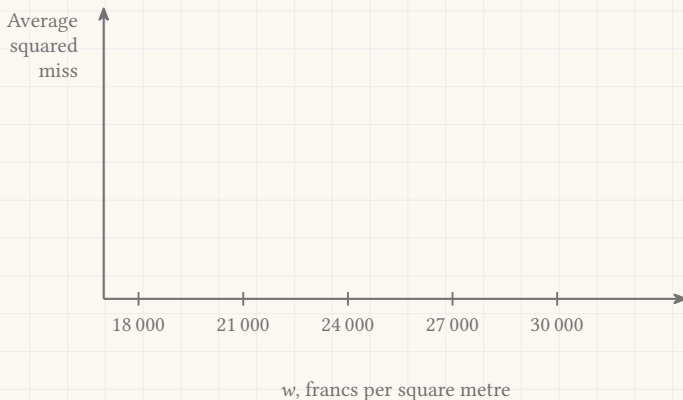
$$\frac{(-500\,000)^2 + (+500\,000)^2}{2} = 500\,000^2$$

Zero only when every miss is zero.

Five houses, one knob

The five houses of the first lecture, priced by area alone. Pick a w , score it, and plot the score.

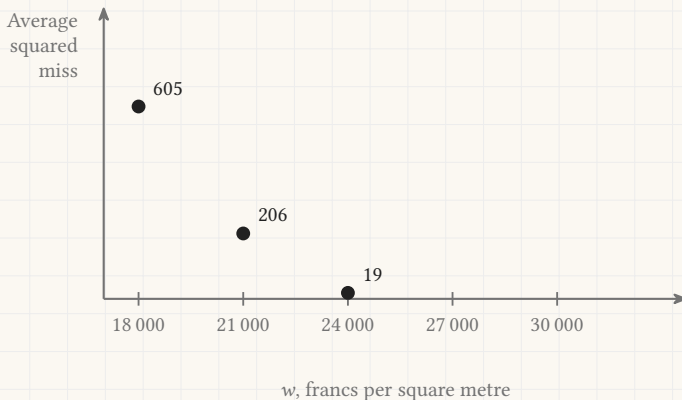
House	Area	Price
1	83	2 156 342
2	62	1 461 635
3	104	2 650 166
4	128	3 184 899
5	145	3 651 113



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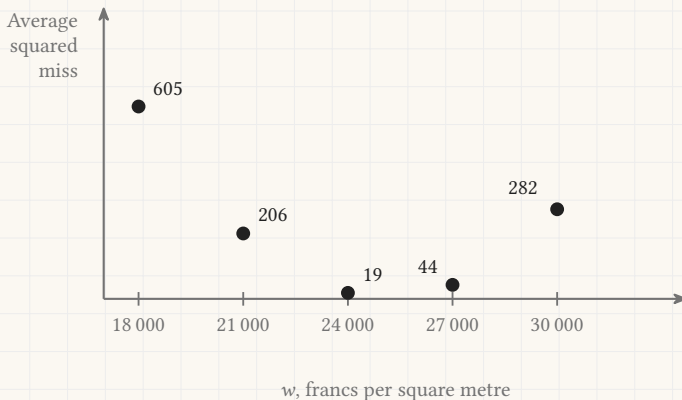
House	Area	Price
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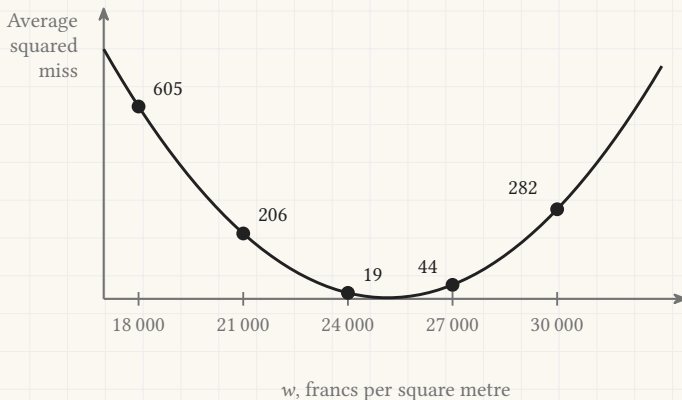
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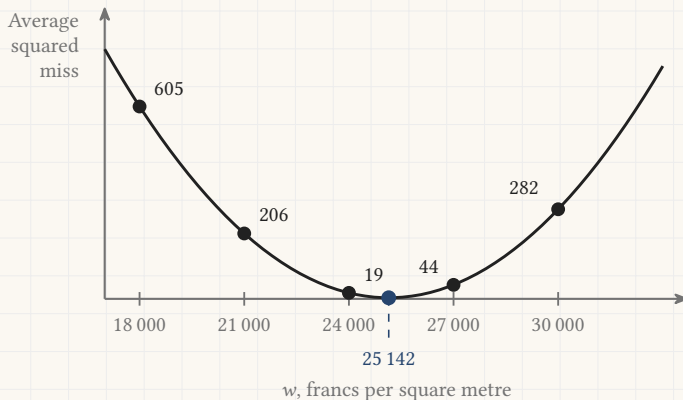
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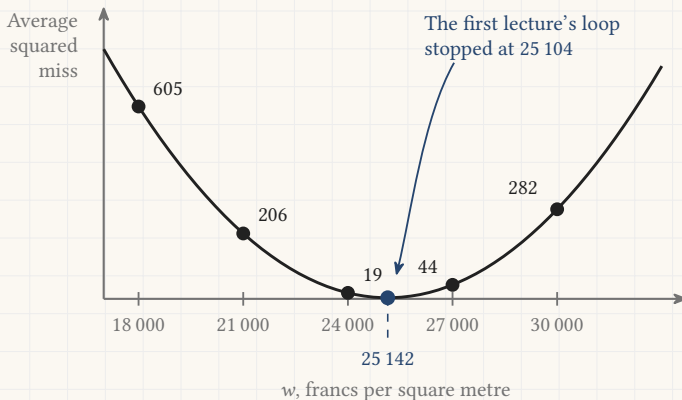
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2

Section 2

Reading a curve

Reading the curve

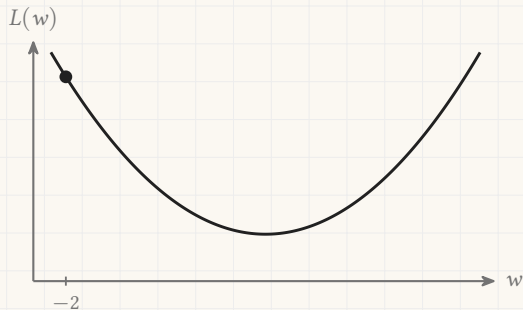
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-----	--------	--------

Reading the curve

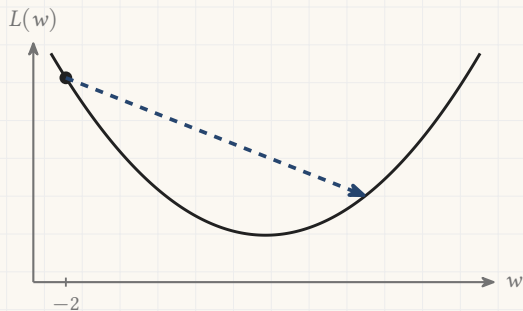
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	

Reading the curve

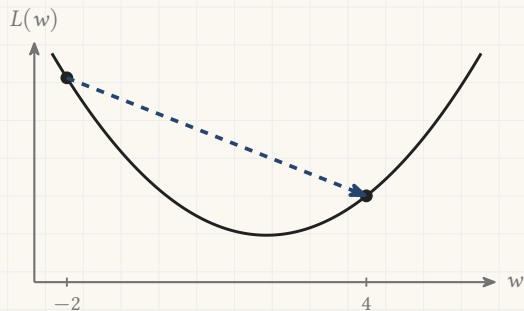
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
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Reading the curve

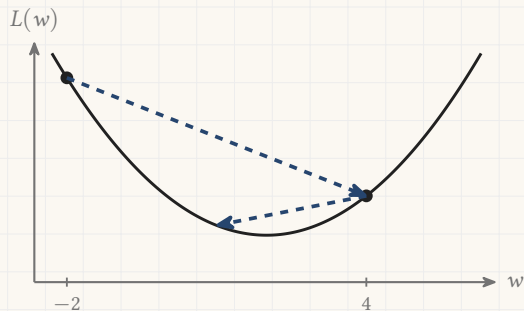
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	
4	7	

Reading the curve

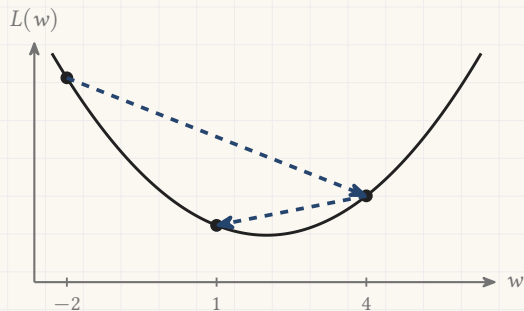
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	
4	7	

Reading the curve

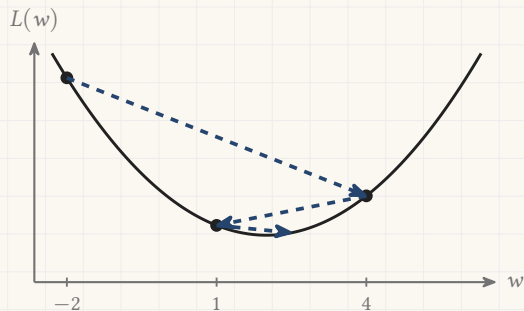
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	→
4	7	←
1	4	

Reading the curve

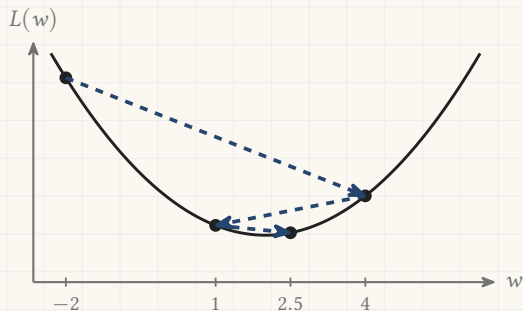
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-2	19	→
4	7	←
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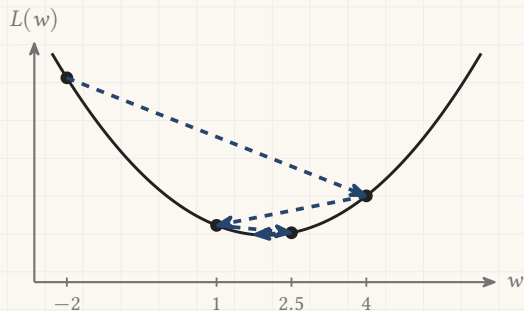
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	→
4	7	←
1	4	→
2.5	3.25	

Reading the curve

Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	→
4	7	←
1	4	→
2.5	3.25	←

Each arrow is half the length of the one above it, and points the other way.

3

Section 3

The grasshopper game

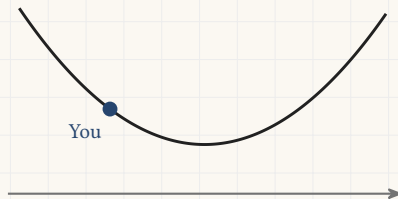
The grasshopper game

You are a grasshopper on a hillside in the dark. You want the lowest point.

Play it here:



https://carloscotrini.github.io/FAI-PAGE/viz/grasshopper/cx_grasshopper.html



The grasshopper game

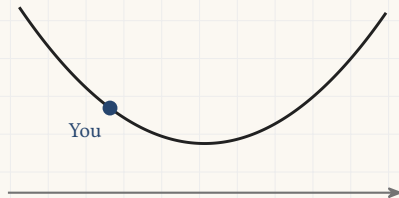
You are a grasshopper on a hillside in the dark. You want the lowest point.

- You know where you are standing.

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The grasshopper game

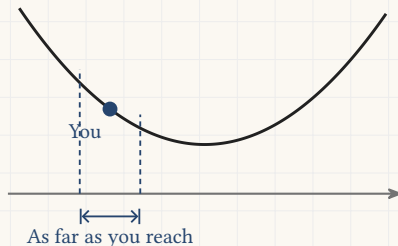
You are a grasshopper on a hillside in the dark. You want the lowest point.

- You know where you are standing.
- You can feel the ground under your feet, as far as your legs reach.

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The grasshopper game

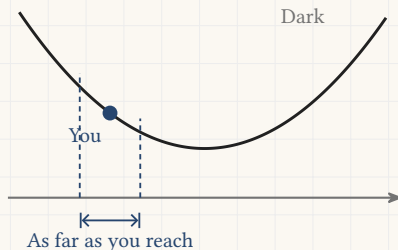
You are a grasshopper on a hillside in the dark. You want the lowest point.

- You know where you are standing.
- You can feel the ground under your feet, as far as your legs reach.
- You cannot see the hillside.

Play it here:



https://carloscotrini.github.io/FAI-PAGE/viz/grasshopper/cx_grasshopper.html



One round of the game

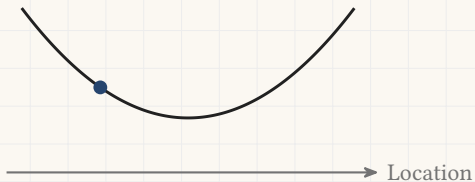
The 09:00 loop again, with the grasshopper's own four words.

Location

Height

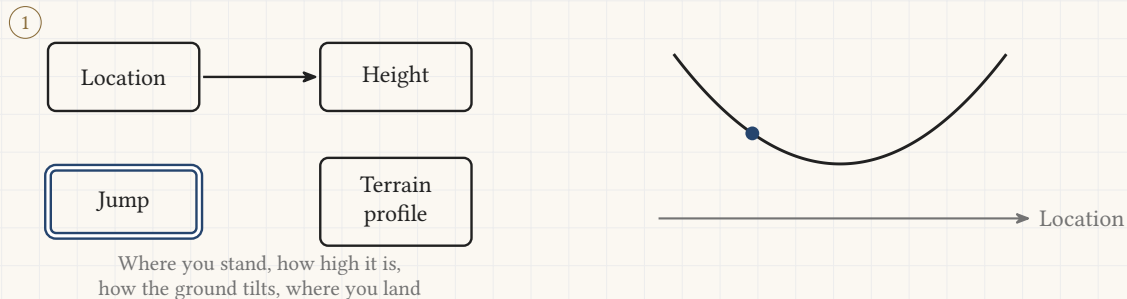
Jump

Terrain profile



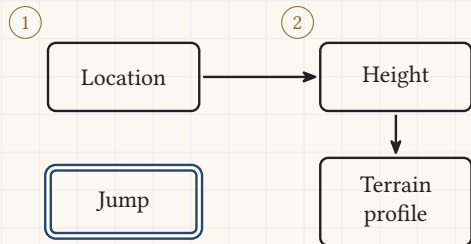
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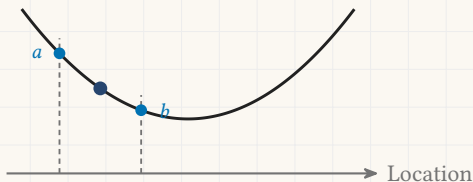


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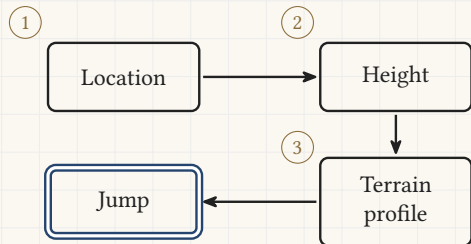


Where you stand, how high it is,
how the ground tilts, where you land

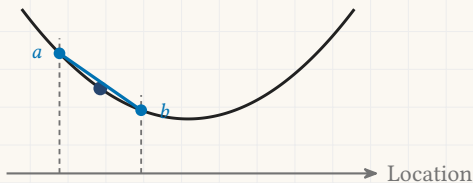


One round of the game

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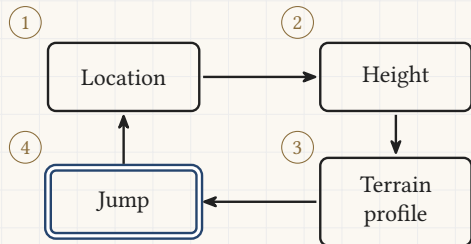
Where you stand, how high it is,
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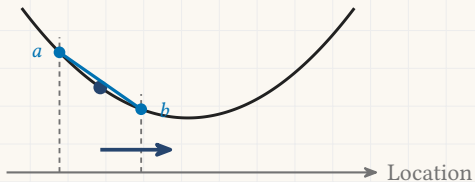
The tilt between the two feet.

One round of the game

The 09:00 loop again, with the grasshopper's own four words.



Where you stand, how high it is,
how the ground tilts, where you land



The tilt between the two feet.

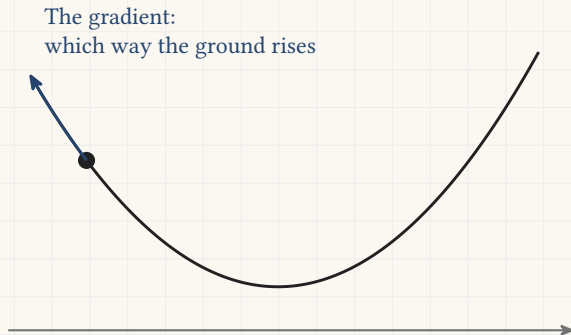
The gradient

Stand anywhere on the hillside. What you can feel underfoot is the tilt of the ground.



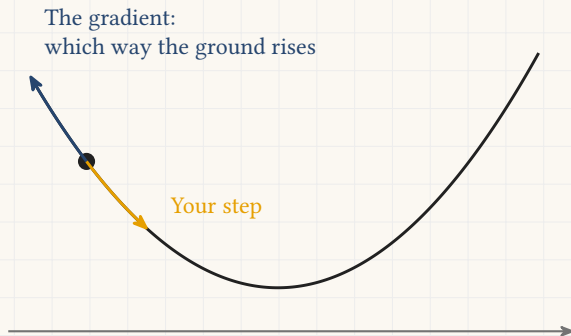
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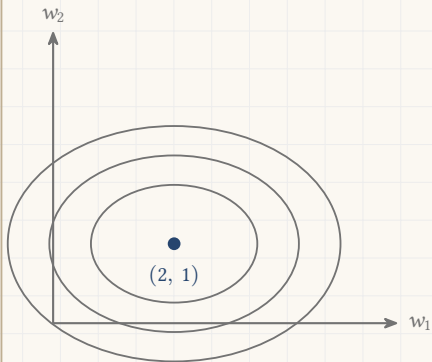
4

Section 4

More than one knob

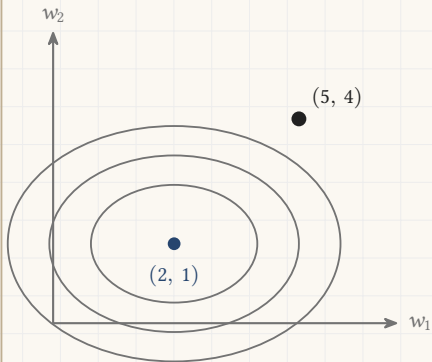
Two knobs

$L(w_1, w_2) = (w_1 - 2)^2 + 2(w_2 - 1)^2 + 3$, seen from above.



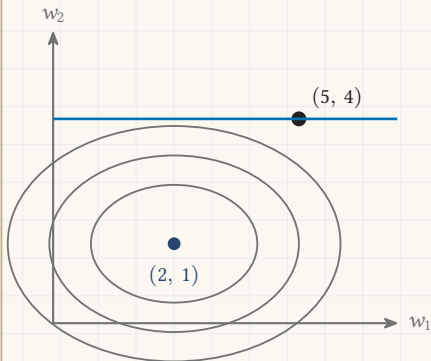
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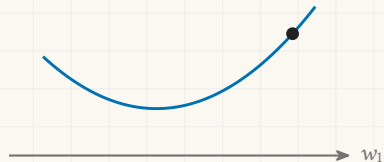


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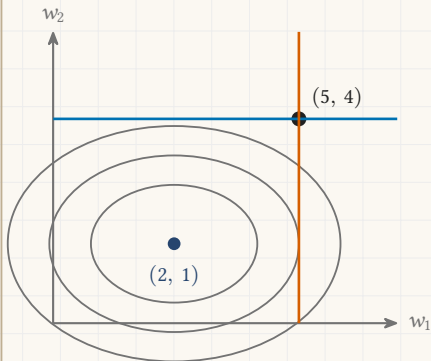


Hold $w_2 = 4$

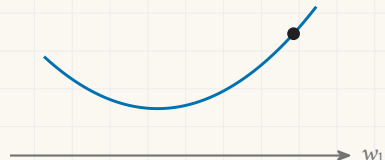


Two knobs

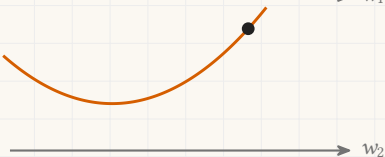
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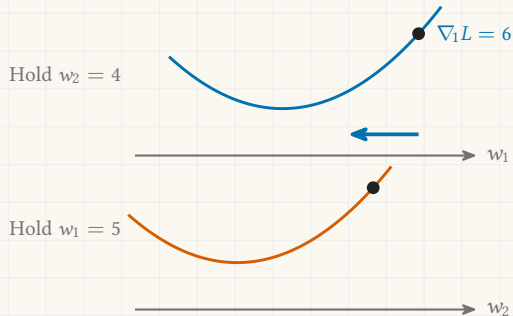
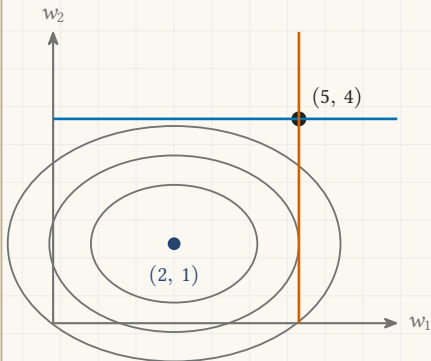


Hold $w_1 = 5$



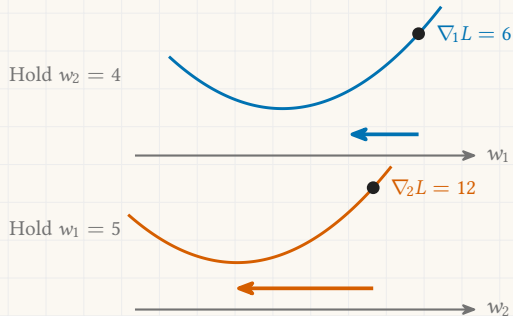
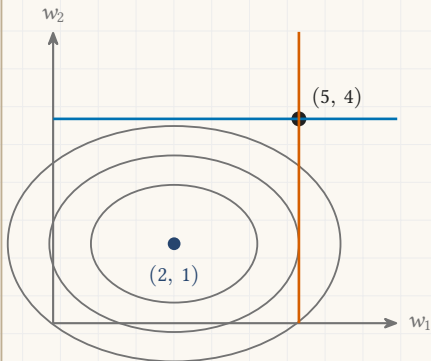
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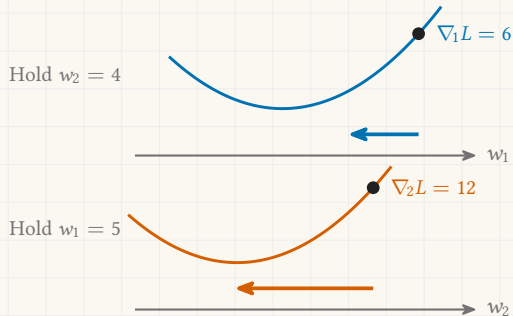
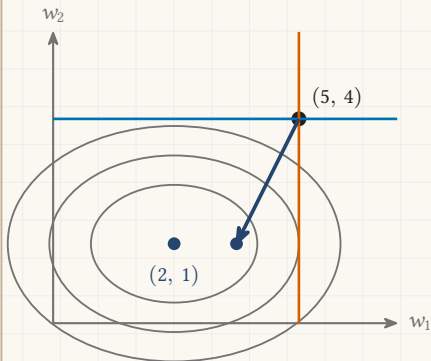
Two knobs

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Two knobs

$L(w_1, w_2) = (w_1 - 2)^2 + 2(w_2 - 1)^2 + 3$, seen from above.



$$\begin{aligned}(w_1, w_2) &\leftarrow (5, 4) - 0.25 \cdot (6, 12) \\ &= (3.5, 1)\end{aligned}$$

Neural networks

Foundations of AI

Carlos Cotrini

UZH CAS Gamechanger AI

26 September 2026

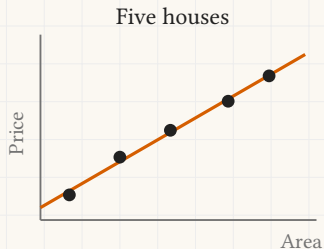
1

Section 1

What a linear model cannot fit

A linear model does not fit everything

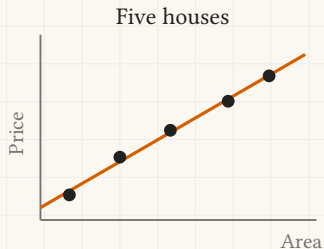
Three sets of measurements, and the best straight line through each.



The largest miss is 83'413 CHF,
which is 3.8 percent of the
price range.

A linear model does not fit everything

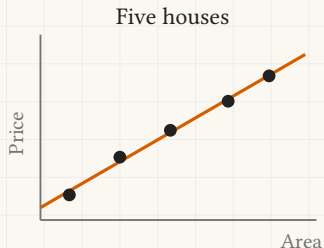
Three sets of measurements, and the best straight line through each.



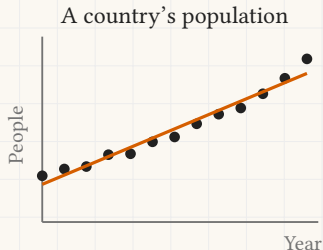
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A linear model does not fit everything

Three sets of measurements, and the best straight line through each.



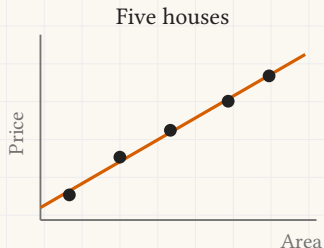
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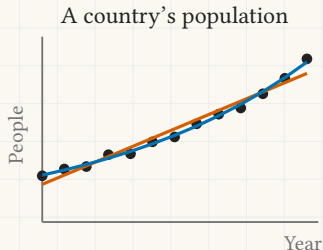
The straight line misses by 12.4 percent of the range. The curve $y = e^{wx}$ misses by 3.2 percent. Synthetic data.

A linear model does not fit everything

Three sets of measurements, and the best straight line through each.



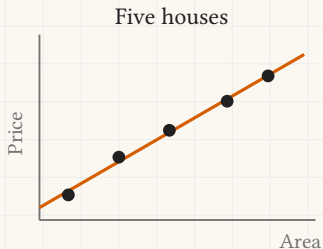
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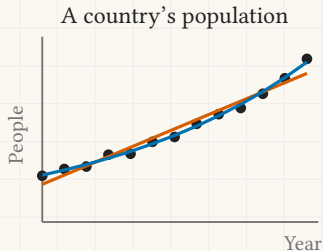
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A linear model does not fit everything

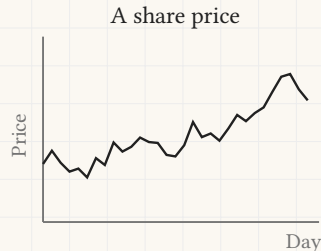
Three sets of measurements, and the best straight line through each.



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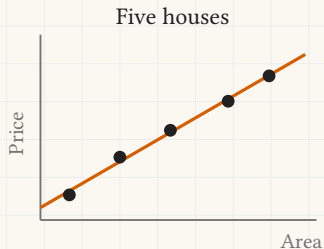


The straight line misses by 12.4 percent of the range. The curve $y = e^{wx}$ misses by 3.2 percent. Synthetic data.

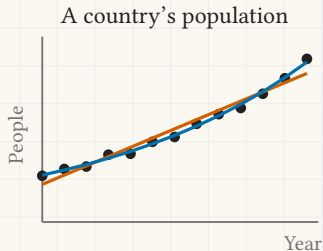


A linear model does not fit everything

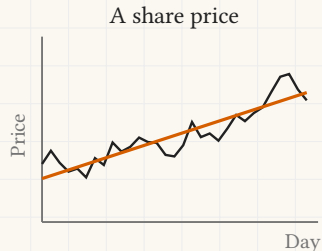
Three sets of measurements, and the best straight line through each.



The largest miss is 83'413 CHF, which is 3.8 percent of the price range.



The straight line misses by 12.4 percent of the range. The curve $y = e^{wx}$ misses by 3.2 percent. Synthetic data.



The straight line misses by 32.8 percent of the range, and no elementary curve does better. Synthetic data.

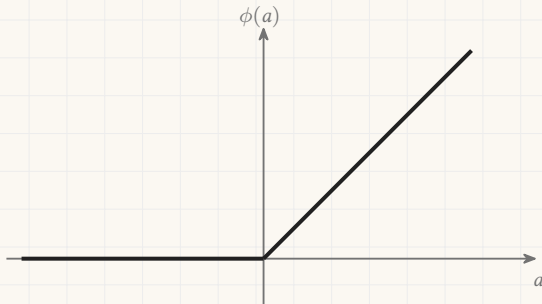
2

Section 2

Neural networks

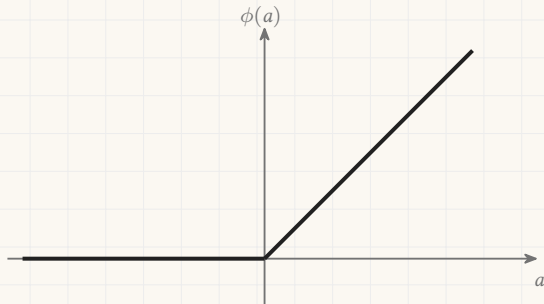
The activation function

ϕ is applied to every entry on its own, and it is the only piece of the network that is not a matrix product.



The activation function

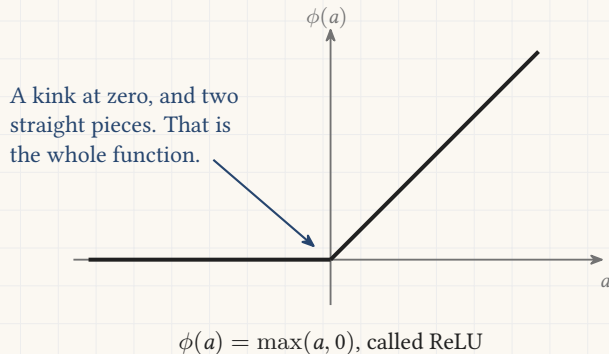
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$$\phi(a) = \max(a, 0), \text{ called ReLU}$$

The activation function

ϕ is applied to every entry on its own, and it is the only piece of the network that is not a matrix product.



The same network, drawn as neurons

One circle for every column of every weight block.

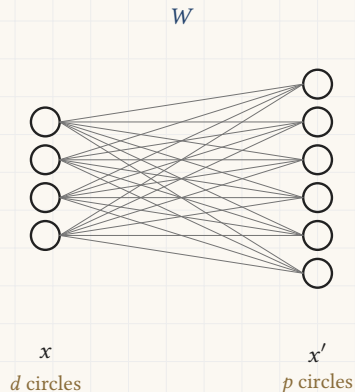


x

d circles

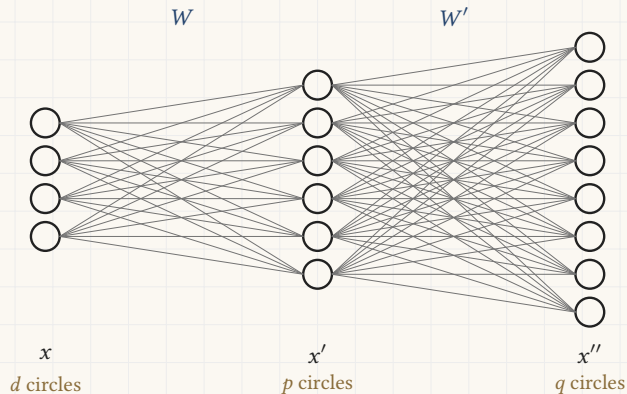
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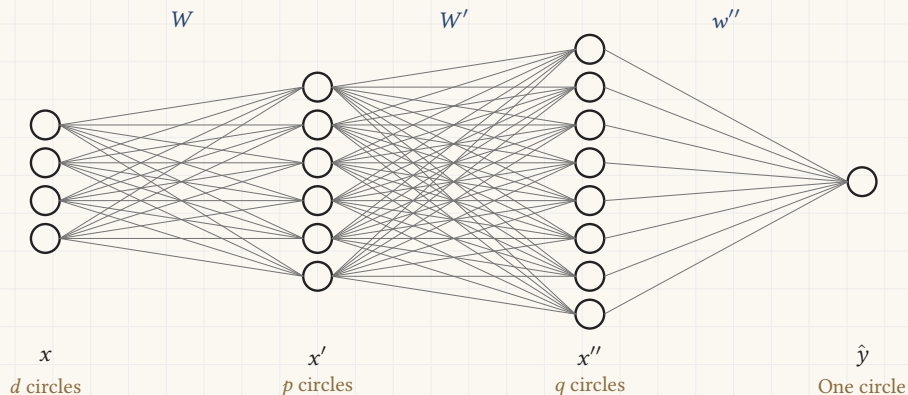
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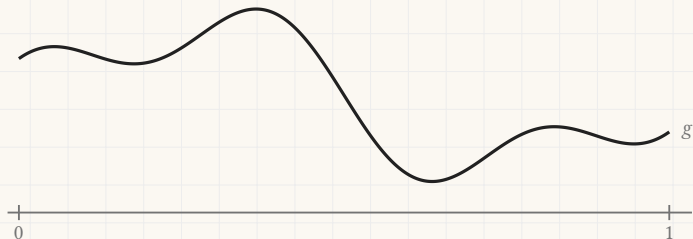
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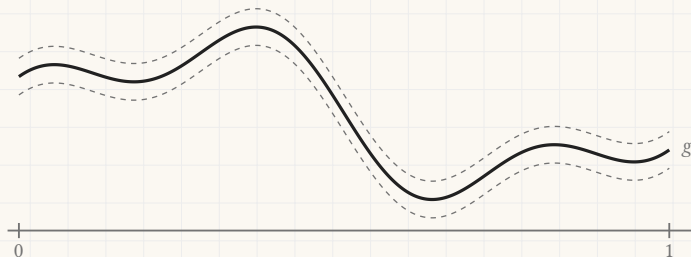
The universal approximation theorem

A continuous target, and networks with p hidden units drawn over it. The band is the accuracy $\varepsilon = 0.25$.



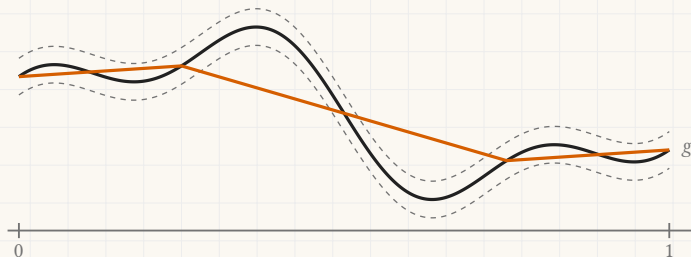
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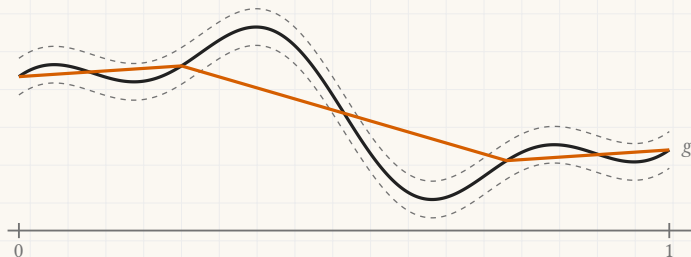
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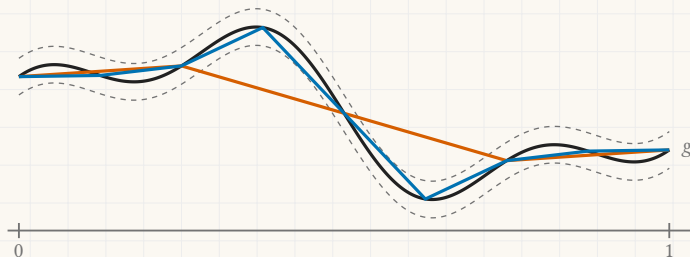
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$p = 4$, largest miss 0.85043

The universal approximation theorem

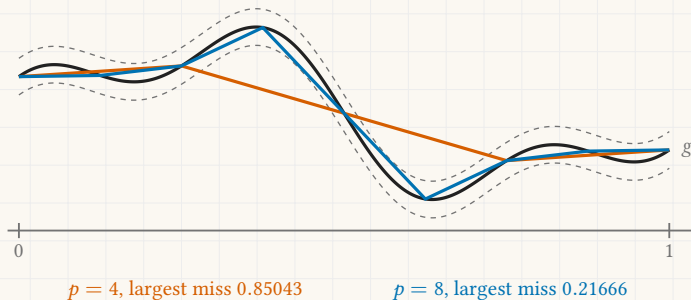
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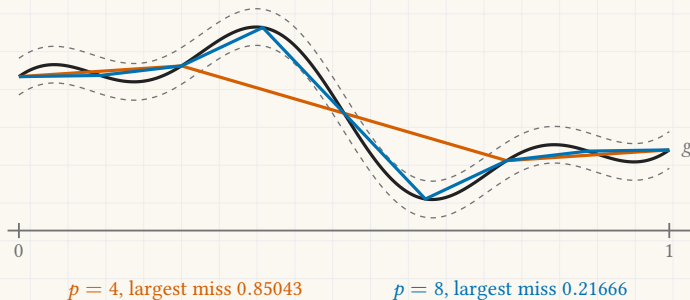
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A network with one hidden layer, and enough units in it, is a universal approximator.

The largest miss falls to 0.06944 at $p = 16$, to 0.01788 at $p = 32$ and to 0.00448 at $p = 64$.

Cybenko, 1989. Hornik, Stinchcombe and White, 1989.

Try it: TensorFlow Playground

Pick the spiral dataset, then add neurons and layers until the two colours separate.

<https://playground.tensorflow.org>

